LoRA-CIM: A Fully-Digital Hybrid Flash-SRAM CIM Accelerator with LoRA Fine-Tuned LLM in Edge Devices

Zhiwang Guo[1], Taoran Shen[1], Jinbei Fang[1], Jingwen Jiang[1], Deyang Chen[1], Hong Nie[2], Jianguo Yang[3], Xiaoyong Xue[1], Xiaoyang Zeng[1]

[1]Fudan University, Shanghai, China; [2]China Flash Co.,Ltd, Shanghai, China; [3]Zhangjiang Laboratory, Shanghai, China.

Email: xuexiaoyong@fudan.edu.cn

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Fine-tuning for task-specific adaptation is usually necessary after extensive pre-training on general-domain datasets for deployment of large language models (LLMs) [1]. Given the power constraint in edge devices, computing-in-memory (CIM) provides a preferable solution for LLM acceleration owing to less data movement between memory and processors. However, deploying LLMs with fine-tuning on CIM-based edge devices faces three critical challenges (Fig. 1): 1) NVMs and SRAM common in present CIM designs both struggle to meet the endurance requirement from fine-tuning and maximum capacity for parameter storage at the same time. 2) Analog CIMs suffer from the accuracy loss due to error accumulation and device non-idealities, as well as high costs in terms of area and power from data conversion circuits. 3) Current CIM designs exhibit low utilization and poor computational efficiency when processing attention and fully-connected (FC) layers in transformers.

This work presents the first fully-digital hybrid Flash-SRAM CIM accelerator scheme, LoRA-CIM, employing low-rank adaptation (LoRA) fine-tuned LLMs on edge devices. 1) The lookup table (LUT)-based Flash CIM engine performs the MAC operations with the pre-trained weights, while the SRAM CIM engine carries out the trainable rank matrices. 2) Fully-digital computation architecture with floating-point supported achieves higher decoding accuracy. 3) Dynamic configuration of attention and FC layers between pipeline and parallelism helps to maximize the utilization of engine resources. LoRA-CIM employs a sub-billion-parameter model MobileLLM-125M [2] with 1Gb weight parameters, which is easily met by state-of-the-art 40nm ETox Flash technology with the secondary electron multiplication injection principle. A 1Gb Flash chip is also given in this work with 0.02um^2 cell size.

Fig. 2 illustrates the hybrid Flash-SRAM CIM structure. The weights of LoRA fine-tuning include pre-trained weights W_0 and trainable rank matrices B and A. LoRA splits the data flow and reduces weight-update parameters in a few MBs. LUT-CIM, based on NOR Flash, calculates MAC of W_0 avoiding weight rewrite power by leveraging the fixed nature of pre-trained weights. The implementation of LUT-CIM consists of two parts: storage and lookup. Firstly, W_0 are multiplied by different IN with external computation units, and then the resulting products are stored in the NOR Flash array. Secondly, LUT-CIM directly readouts the products through array addressing with sense amplifiers (SAs). The LUT-CIM NOR Flash reduces 55% MAC compute latency and 75% power compared with NVM-based analog CIM. Moreover, it also achieves the highest density and smallest area in recent works. The SRAM CIM engine performs MAC operations of trainable rank matrices and updates the LoRA matrices during fine-tuning. The hybrid Flash-SRAM CIM structure features large capacity, infinite endurance, and rapid response characteristics for fine-tuned LLMs with reductions of 95% write duration time and 98.2% write energy compared with NVM-based analog CIM.

Fig. 3 shows the overall architecture of LoRA-CIM with floating-point supported. LoRA-CIM consists of 64 CIM cores, and each CIM core includes 3 NOR Flash processing elements (PEs), a 64kb input buffer, a 128kb global buffer, 2 SRAM PEs, and adder-tree. Each Flash PE contains 4 Flash banks Q, K, V, and O, each of which includes 80 Flash engines. The NOR Flash engine (64kb) comprises 64 Flash CIM subarrays, WL, SL, BL decoders and drivers, and 40 SAs shared across the 64 subarrays. Each subarray contains 32 Flash multiplier units (FMUs), the FMUs in each column work cooperatively. The FMU structure of 2-bit IN × 8-bit W achieves balance between area of LUT units and power consumption of SAs. Each SRAM PE consists of 32 SRAM engines. Additionally, the SRAM engine (4kb) contains 64 SRAM multiplier units (SMUs), decoders and drivers, and adder-tree. Moreover, FMU and SMU can also be compatible with floating-point operations. BF16 multiplication requires 4 computation cycles using 2 FMUs or 2 SMUs. The 1-bit sign and 8-bit exponent are directly stored in one FMU, with the same 9-bit value stored across four WLs in the FMU. A 1-bit XOR gate and adders handle the computation of the sign and exponent, while another FMU executes the multiplication of the 8-bit mantissa (7-bit mantissa and 1-bit hidden bit). Similarly, SMU follows the same computation process for BF16. The fully-digital architecture based on LUT-CIM without accuracy loss under BF16 precision. LoRA-CIM enables to approach the accuracy of LLM software implementations.

Fig. 4 illustrates the pipeline and parallelism modes for transformer block. For attention and FC layers, LoRA-CIM improves energy efficiency through dynamic configuration of attention and FC layers between pipeline and parallelism. The process of fine-tuning can be divided into three stages. Firstly, the weight load stage writes weights of NOR Flash and SRAM. Secondly, in the decoding stage, MAC operations mainly exist in attention layers and FC layers (linear layers) of transformer blocks. For attention layers, LoRA-CIM is configured in a pipeline mode, directly streaming the 1st-stage CIM core to the 2nd-stage CIM core. In CIM core0 for pipeline1, Flash PE0 stores the weight matrix W_0, and the SRAM PE0 stores the weight matrix BA of LoRA. The input buffer of the core0 provides token embedding inputs x to both Flash PE0 and SRAM PE0, enabling the generation of W_0x and BAx. They are inputted into the adder-tree to calculate the embedding results h. Besides, the embedding results h will be inputted from global buffer to the Flash PE1 and Flash PE2, which compute the K and Q respectively. In CIM core1 for pipeline2, Flash PE0 stores the weight matrix W_V, and calculates the W_Vh to generate V. SRAM PE0 in two CIM cores stores the LoRA weight matrix BA and calculates BAx. Meanwhile, Flash PE1 stores the weight matrix K^T from the Flash PE1 in CIM core0, and multiples K^T by Q to compute A. The adder-tree in CIM core1 is bypassed. The Flash PE2 stores the weight V, and compute A'V getting the attention layer output. For FC layers, LoRA-CIM will work in a parallelism mode, whose CIM cores work independently and perform MAC operations of pre-trained weights, while SRAM PEs are bypassed. Thirdly, the fine-tuning stage contains backpropagation (BP) process and weight update (WU) process. In BP process, LoRA-CIM works in a parallelism mode. Three Flash PEs in two CIM cores calculate the product of $\partial L/\partial h$ and W^T, and SRAM PEs work for the product of $\partial L/\partial h$ and BA. Finally, in WU process, to be updated BA will be written in SRAM engine, while W remains unchanged. The pipeline and parallelism for transformer block improves up to 1.75x and 1.5x energy efficiency compared with the Flash baseline and SRAM baseline, respectively.

Fig. 5 shows the measurement results of LoRA-CIM. LoRA-CIM achieves promising ~5% improvement of few-shot common-sense reasoning accuracy compared to CIM without fine-tuning. Moreover, MobileLLM-125M with transformer sharing block compresses parameters into 1Gb, making it more suitable for LLM implementation on edge devices. In conclusion, LoRA-CIM achieves 356tokens/s and 6tokens/s decoding speed for MobileLLM-125M and Llama2-7B. It represents 1.9x/7.7x and 2x/6x decoding speed improvement of the MobileLLM-125M/Llama2-7B compared to the Flash baseline and SRAM baseline. MobileLLM-125M and Llama2-7B require 86.2uJ/token and 4.83mJ/token for LoRA-CIM decoding, which can reduce 43% and 40% energy consumption at most.

Fig. 6 shows a comparison with the state-of-the-art AI hybrid CIM chips. LoRA-CIM is the first fully-digital hybrid test chip that targets LLMs and fine-tuning while supporting floating-point with 1.45TOPS/W and 700GFLOPS/W peak energy efficiency for INT8 and BF16. It features the smallest cell area, ultra-low power consumption, and high energy efficiency. Fig. 7 shows the die micrograph on entire wafer and the chip summary.

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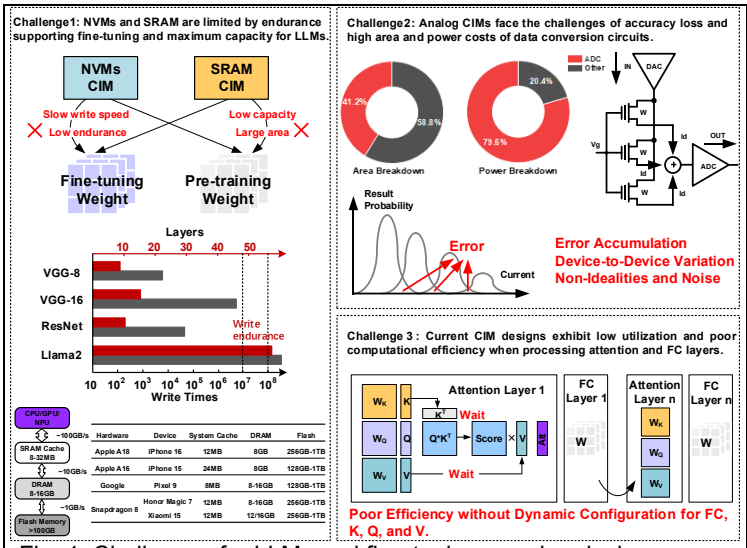


Fig. 1. Challenges for LLMs and fine-tuning on edge devices.

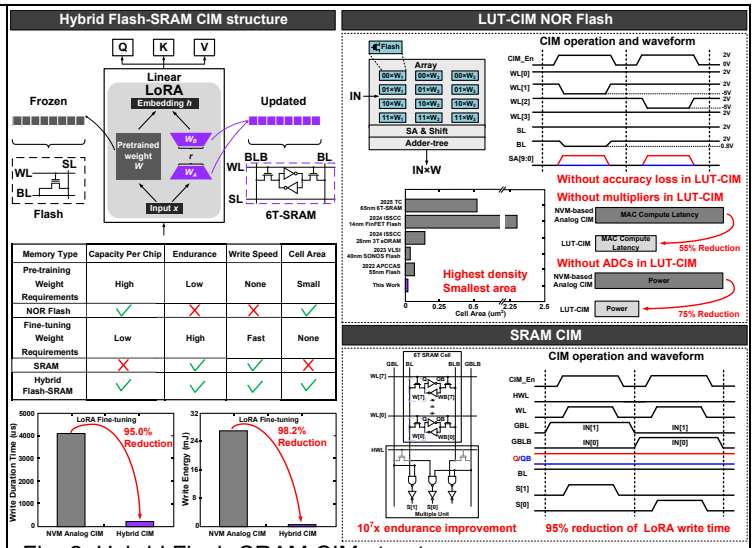


Fig. 2. Hybrid Flash-SRAM CIM structure.

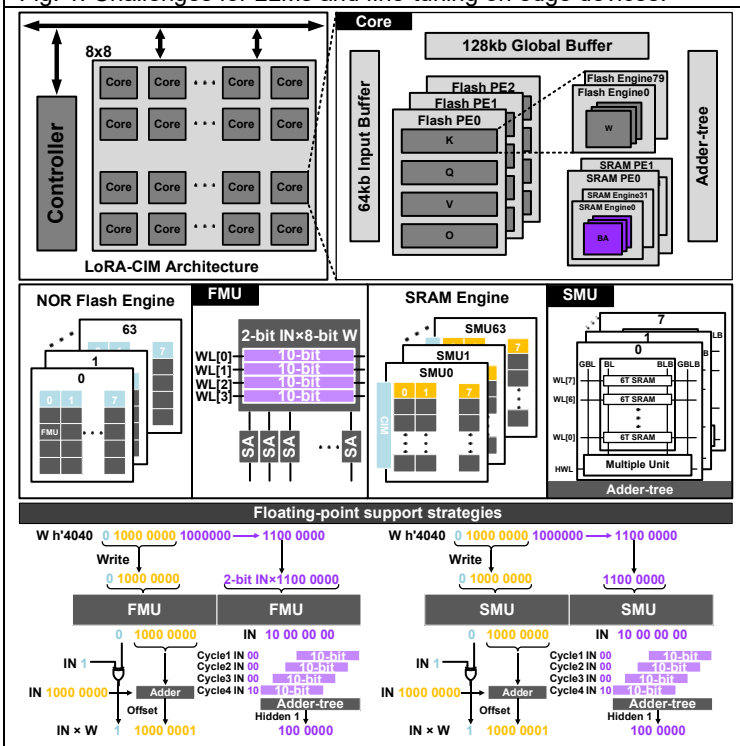


Fig. 3. Overall architecture of LoRA-CIM with floating-point supported.

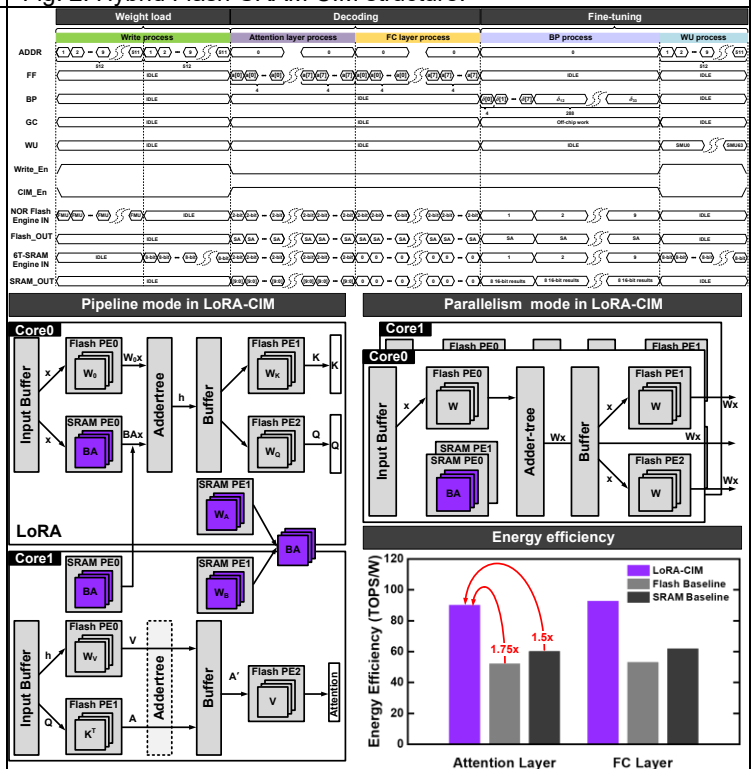


Fig. 4. Dynamic configuration of attention and FC layers.

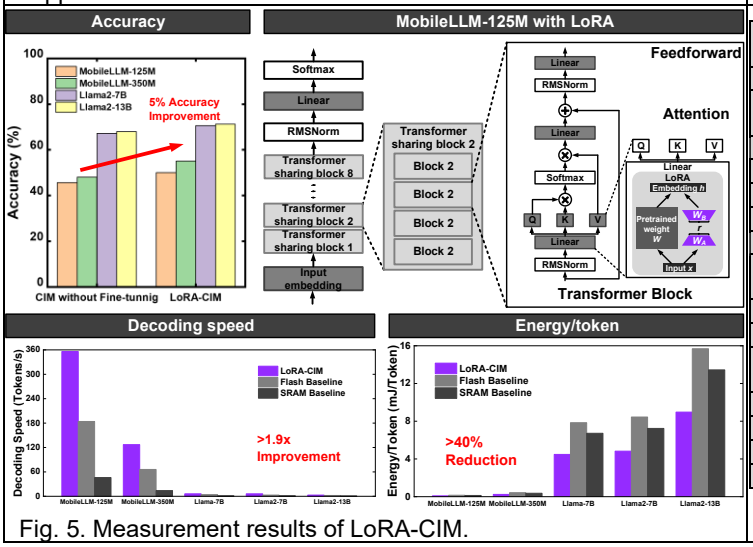


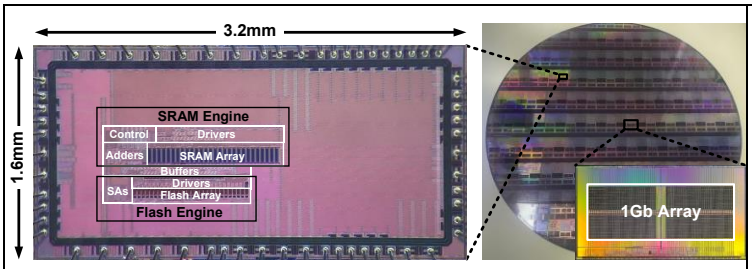
Fig. 5. Measurement results of LoRA-CIM.

	ASP-DAC '20 [3]	ISSCC '23 [4]	ISSCC '23 [5]	ASSCC '24 [6]	ISSCC '23 [7]	This Work
Memory Type	ReRAM+Cap	ReRAM+SRAM	SRAM	ReRAM+SRAM	SRAM	Flash+SRAM
Bit-cell	1T1R Cap	1T1R 6T- SRAM	8T- SRAM 10T- SRAM	1T1R 6T- SRAM	6T- SRAM	NOR Flash 6T- SRAM
Cell Size (um^2)	0.26	0.114	0.739	0.066	0.1	**0.02**
Technology	32nm	40nm	28nm	28nm	12nm	40nm
Normalized Cell Size (F^2)	253.8	71.2	942.6	84.2	694.4	**12.5**
Computing Approach	Analog CIM	Analog CIM	Digital CIM	Digital CIM	Digital CIM	Digital LUT-CIM
Power (mW)	79.72	21.3	8.27	71.95	9 (FP4)/10 (FP8)	**1.04**
Precision (Input, Weight)*1	INT (1,8)	INT1 (1,1)	INT8 (8,8)	INT4/INT7 (4,4)/(7,7)	FP4/FP8 (4,4)/(8,8)	INT8/BF16 (8,8)/(16,16)
Energy Efficiency (TOPS/W)	3.83	73.53	1.76	22.07 (INT4,INT4)	N.A.	1.45 (INT8,INT8)
FOM*2 (TOPS/W)	24.51	73.53	78.82	247.18	N.A.	**92.8**
Energy Efficiency (TFLOPS/W)	N.A.	N.A.	N.A.	N.A.	18.1 (FP4,FP4) 8.24 (FP8,FP8)	0.7 (BF16, BF16)
FOM*2 (TFLOPS/W)	N.A.	N.A.	N.A.	N.A.	86.88	**179.2**
Endurance Friendly	No	No	No	No	No	Yes
Transformer	No	No	Yes	Yes	Yes	Yes
Fine-Tuning	No	No	No	Yes	No	**LoRA**

*1 Precision (Input, Weight) = INT/FP/BF (Input-bit, Weight-bit)
*2 FOM = Input-precision × Weight-precision × Energy Efficiency and scaled to the same technology.

Fig. 6. Comparison with the state-of-the-art AI hybrid CIM chips.

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Technology	40nm ETox Flash
CIM Size	64kb Flash + 4kb SRAM
Die Area	3.2×1.6mm²
Supply Voltage	2.0V
Frequency	50MHz
Precision	INT8, BF16
Power	1.04mW
Energy Efficiency	1.45 TOPS/W(INT8) 0.7T FLOPS/W(BF16)
Energy/Token	86.2uJ/token(MobileLLM-125M) 4.83mJ/token(Llama2-7B)

Fig. 7. Die micrograph and chip summary.

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